NIKUNJ GARG

Java Backend Developer | BCA Student

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SUMMARY

BCA student with hands-on experience developing high-concurrency backend systems using **Java 21**, **Spring Boot 3**, **PostgreSQL**, and **Redis**. Built systems involving **distributed locks (Redisson)**, ACID concurrency control, idempotency gates, stateless **JWT** security, and real-time **WebSocket (STOMP)** messaging. Backed by automated **JUnit 5** stress and failure-mode testing.

TECHNICAL SKILLS

- **Languages:** Java 21, SQL
- **Backend & Distributed:** Spring Boot 3, Spring Data JPA, Hibernate ORM, RESTful APIs, WebSockets (STOMP), Distributed Locks (Redisson)
- **Databases & Caching:** PostgreSQL, Redis 7 (TTL Leases, Keyspace Events), Schema Design (3NF), ACID Transactions, Locking (Pessimistic/Optimistic)
- **Security & Testing:** Spring Security, JWT (Refresh Token Rotation), Role-Based Access Control (RBAC), JUnit 5 (250+ Tests, Concurrency Suites), Mockito
- **Tools & DevOps:** Docker, Docker Compose, Git, GitHub, Maven, Postman, Micrometer, OpenAPI 3.0, Render

PROJECTS

DevTinder — Developer Matching & Messaging Platform | Java 21, Spring Boot 3, PostgreSQL, WebSockets (STOMP), Spring Security, Docker GitHub Repo ↗

- Built bidirectional real-time messaging using **Spring WebSocket (STOMP)** paired with REST delta synchronization (`?after={id}`) and PostgreSQL persistence so users can retrieve missed messages upon reconnecting.
- Implemented a developer discovery and matching service that ranks candidate profiles dynamically by shared skill counts in descending order and tracks mutual interest using unique composite constraints on swipe records.
- Hardened authentication using stateless **Spring Security** filters, short-lived **JWT** access tokens, sliding-window IP rate limiting, and **Refresh Token Rotation (RTR)** with token family tracking to detect and revoke compromised sessions.
- Containerized the backend using a multi-stage **Dockerfile** and configured environment variables and health-check endpoints (`/api/health`) for cloud deployment integrating with a Next.js frontend.

OpenBounty — High-Concurrency Challenge & Milestone Engine | Java 21, Spring Boot 3, PostgreSQL, Spring Data JPA, Spring Security, JUnit 5 GitHub Repo ↗

- Engineered an enterprise bounty marketplace with atomic proposal acceptance using **pessimistic row-level locking** (`SELECT ... FOR UPDATE`), serializing competing bids with zero lost updates.
- Designed a **multi-stage deliverable verification** state machine with sequential order guards, IDOR client protection, and auto-completion algorithmic reputation triggers (`+20 pts`).
- Hardened data integrity against **double-submission race conditions** using **pessimistic write locks** on user rows and case-insensitive active-bounty uniqueness guards.
- Built a **250+ automated test suite** in **JUnit 5**, including multithreaded concurrency tests (`CountDownLatch` / `ExecutorService`) and adversarial penetration suites verifying zero bypass under load.

TicketWave — Distributed Seat Hold & Checkout Engine | Java 21, Spring Boot 3, Redis 7 (Redisson), PostgreSQL, Micrometer, JUnit 5 GitHub Repo ↗

- Architected distributed seat hold allocation using **Redisson MultiLock** across Redis (`<1ms`) with deterministic seat-ID sorting to eliminate deadlocks under high-concurrency arena ticket spikes.
- Built a **Redis SETNX idempotency gate** (`IN_PROGRESS` 120s / `COMPLETED` 24h) and decoupled payment execution outside DB boundaries, preventing HikariCP pool exhaustion with atomic PostgreSQL commits.
- Engineered a **dual-tier inventory reclamation** pipeline (Redis keyspace pub/sub + scheduled PostgreSQL sweeper) to auto-reclaim expired 10m holds; verified zero overselling with **70 automated JUnit 5 tests**.

EDUCATION

Bachelor of Computer Applications (BCA) — Shri Guru Ram Rai University, Dehradun **2025 – 2028**

Cumulative CGPA: 7.48 / 10.0 • **Semester 2 SGPA:** 7.96 / 10.0

LEADERSHIP & ACHIEVEMENTS

- **Secretary — TechMantra Coding Club:** Promoted from Core Member to Secretary within 10 months; organized 5+ hands-on coding bootcamps and technical workshops for 120+ student developers.
- **Startup Innovation Recognition:** Awarded Certificate of Recognition from **UCOST & SGRR University IIC** for pitching the startup concept *College Clash* in university innovation competitions.
- **Hackathons & Competitions:** Engineered and deployed DevTinder backend in 72 hours during **Devlynix Buildathon 2.0**; awarded Certificates of Participation in **Algothon'26 (Algoxilla)** and **Smart India Hackathon (SIH'26 Internal Round)**.
- **Algorithmic Problem Solving:** Solved data structures and algorithms problems in Java on LeetCode (@Nikunjgarg12), focusing on arrays, strings, two-pointers, and linked lists.